Seminar "Internet Measurement"

Live Streaming Performance of the Zattoo Network

Fakultät IV
Technische Universität Berlin

eingereicht von:
Michael Winkelmann (338024)
(miwinkel@mailbox.tu-berlin.de)

Sommersemester 2011

Abstract

The Zattoo network is a P2P live streaming system. To improve the quality of service for such systems, a variety of measurements need to be performed to detect weaknesses and bottlenecks. Due to their decentral nature, P2P systems behave differently than conventional server-based systems and thus, also the measurement methology differs. In paper [3], the Zattoo Live Streaming Network was examined. This term paper describes how these measurements in this study were performed and what observations have been made. The measurements for sharing ratio and NAT traversal will be analyzed in more detail. Moreover, the measurements and observations will be compared to those performed in other studies. Furthermore, the results will be discussed and an outlook for further improvements in measurement methology as well as P2P-based live streaming is given.

1 Introduction

With the growing availability of broadband connections and the gaining success of web platforms like YouTube and Vimeo, digital video streaming over the internet is becoming more and more popular and the quality and variety of multimedia and TV streaming services are growing rapidly [4].
One kind of media streaming is peer-to-peer streaming (P2P streaming). In contrast to client-server-based streaming, a P2P system consists of a set of peers arranged in a decentral manner. These peers are not only receiver of media data, they also send the currently buffered media to other available receivers. Due to its decentral nature, P2P can provide a higher reliablity and decreased operational costs compared to client-server architectures [14]. For a better understanding and for further improvements of P2P live streaming systems, a variety of measurements needs to be performed.
Because the majority of P2P streaming solution are proprietary, most of the studies which examine the behaviour of such systems only consider the packet traces from end-user perspective and thus, the observed behaviour is not always intuively understandable. For more elaborate analyses, costly reverse engineering is often required.
A lot of commercial P2P-based systems for live streaming like Zattoo, PPlive, Joost, LiveStation, SOPCast, TVants have emerged in recent years [6]. With three million registered users in eight European contries, Zattoo is the largest production live streaming providers in Europe. The system rebroadcasts TV channels received from satellites to the internet. The papers [3; 14] analyze Zattoo network extensively. The measurements in this study

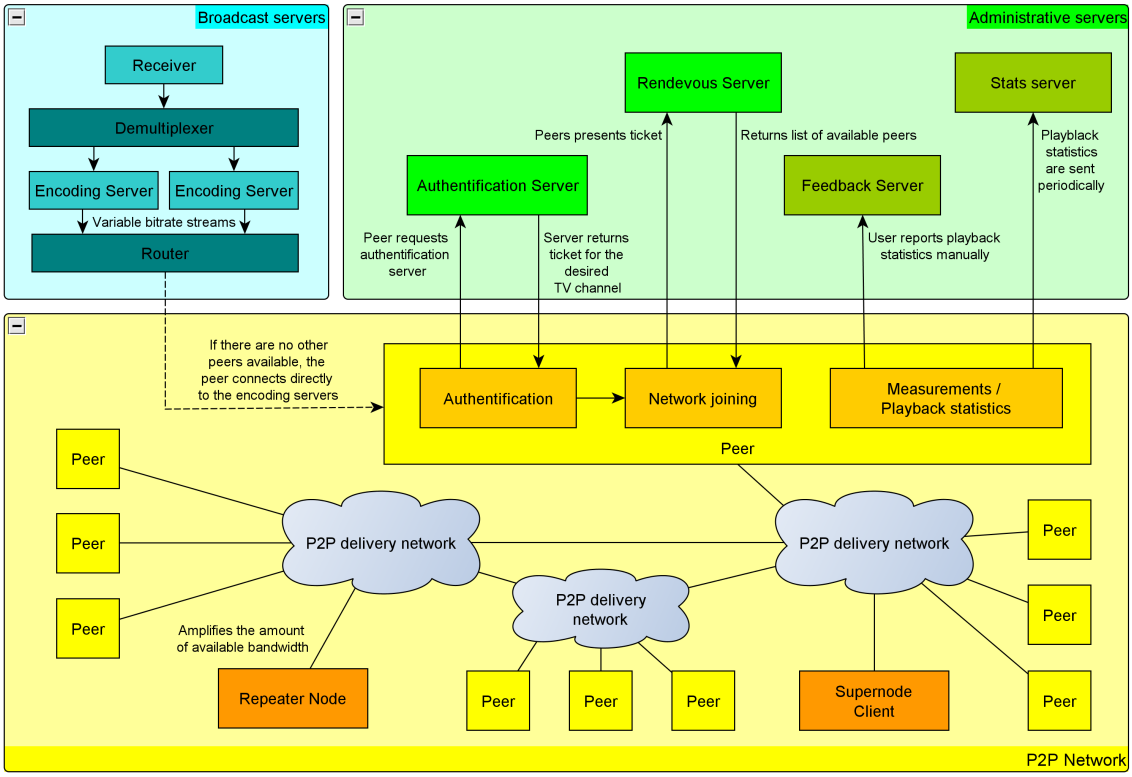

Figure 1: Overview of the Zattoo live streaming system.

are based on usage statistics and error reports collected from the provider.
This term paper will describe the methology and the obtained results of measurements in a P2P-based system with focus on the Zattoo system. The measurements for NAT traversal and sharing ratio will be analyzed in more detail. The text is organized as follows: The functionality of P2P-based streaming systems will be described examplarily on the Zattoo system. Afterwards, the measurements conducted in the studies [3; 14] will be explained. The paper is finalized with a conclusion and a discussion about further improvements of a P2P live streaming system.

2 Functionality of P2P-based real-time live streaming

A P2P live streaming system contains several different elements which act together. An overview of the different components of the Zattoo system is shown in figure 1.
These elements can be grouped into *broadcast servers* (blue in figure 1), *administrative servers* (green) and the *P2P network* (yellow).

2.1 Broadcast servers

The broadcast servers (depicted light blue in figure 1) are responsible for receiving, encoding the media signals and finally, to transfer them into the P2P network. In the P2P network, each peer can be seen as different user watching one TV channel at a time.

Receivers

The receiver may capture the media data from certain external sources like satellites, cable TV or DVB-T. The Zattoo system receives live TV signals captured from satellites. The

different received TV channels are demultiplexed and sent to the encoding servers.

Encoding servers

The encoding servers get different TV channels from the receivers and encode them into several H.264/AAC variable bit-rate streams. Furthermore, an encryption with for the streams may be applied [3]. Each live stream is sent out as a packet stream with 16 logical sub-streams. A sub-stream is subdivided into segments with a fixed duration of about a second [1].
If a certain peer wants to receive a specific live stream and no other peers are reachable or the network capacities of the P2P delivery network are limited, a peer connects directly to an encoding server to receive the desired live stream.

2.2 Administrative servers

There are several types of administrative servers controlled by the provider required for authentification and measurements for playback and network conditions.

Authentification server

If a user wants to watch a certain channel, he has to request the authentification server where he gets a ticket for the TV channel of interest with limited life time. After the authentification, the user presents his ticket to the rendezvous server.

Rendezvous server

If the user is authorized, then joins peer-to-peer network with this channel and the rendezvous server returns a list of available peers. If this list is empty, this means that user is first user of this channel and thus he will be directly connected to an encoding server.
If the list contains at least one peer, the receiving peer will contact these peers in the list and specifies the sub-streams it would like to receive. The receiving peer decides how to multiplex a stream onto its set of neighboring peers.

Stats and feedback server

For exact measurements, a system needs to be monitored continously and the data needs to be recorded, mostly in form of protocols which may contain data about network and playback conditions. Additionally, the user shall be able to send feedback, for example if an error occurs or the the quality of the video playback is not sufficient.
For this purpose, usually there are two kinds servers in a P2P-system for collecting and processing the usage statistics and error reports: "stats" servers and "user feedback" servers [3; 10; 14]. The "stats"-server periodically collects player statistics from individual peers and aggregates them into full session log files. These log files are entered into a database. In the Zattoo system, the connection and playback statistics are sent every 10 minutes to the server whereas in the Spotify music-on-demand system, the send interval is about 30 minutes [10].

2.3 P2P Network

The P2P network is the major part of the system. It consists of several P2P delivery networks to which at least one peer is connected. Furthermore, network contain repeater nodes and super-nodes if the overall upload bandwidth of the peers is not sufficient.

P2P delivery networks

In Zattoo, every transmitted TV channel is in a separate P2P delivery network. The network is not limited in the number of TV channels it can carry, but a user can only watch one channel at a time. When a user performs a switch from one TV channel to another, he will connect to a different delivery network.

Peer-division multiplexing

The Zattoo system uses a special live streaming protocol which is a receiver based, peer-division multiplexing protocol (PDM). When a peer joins a TV channel, it establishes a peer-division multiplexing scheme amongst a set of neighboring peers. The PDM process consists of a search phase and a join phase [3].
The search phase is initiated by sending out a SEARCH message to all existing peers which are in the list of randomly selected peers returned by the rendezvous server. The SEARCH message contains information about the required sub-stream segments. The existing peers respond to the SEARCH message with number of sub-streams they can forward and number of random set of their neighbor peers. Furthermore, the SEARCH message contains packet numbers of the sub-streams which will be later used for synchrony measurements.
Based on this information, the joining peer waits until a full set of sub-streams for the selected TV channel is available and generates a list of potential neighbor peers. If a joining peer cannot form a set of potential neighbors that covers all of the sub-streams, more SEARCH messages are sent to previously learned neighbor peers.
Another criterion for selecting a potential neighbor is its topological location which is defined by its subnet number, autonomous system number or country code. The topological location from a joining peer is obtained during the authentication process. By considering the topological location, a topology-aware overlay network is build which minimizes the number of physical links and hops between the peers to enhance the quality of service.
After the list of potential neighbors has been determined, the joining peer sends JOIN requests to each potential neighbor. The JOIN request contains the sub-streams the joining peers wants to receive. When a JOIN request was acknowledged, the download of the desired sub-streamed segments starts.

Super nodes and repeater nodes

Because the upload bandwidth of a peer is usually much lower than the download bandwidth, the system often contains several super nodes which increase the available upload bandwidth and reliability. Furthermore, the super nodes are required to assure the scalability of the network.
Repeater nodes amplify the overall amount of available bandwidth in the network whereas the super nodes are responsible to keep a balance between upload and download bandwidth.

3 Measurements in the Zattoo Network

For extensive and representative measurements, large datasets are required. Such a dataset may consist of several kinds of protocols logging different data and events.
Session logs and user feedback reports in studies [3; 14] were recorded during the UEFA Euro Cup in June 2008. Approximately 4500 feedback logs and 7000 session logs from the TV channels ARD (DE), Cuatro (ES) and SF2 (CH) with 5 million sessions of more than a half million distinct users were collected.
The network usage varies strongly over the day and thus, the 24-hour day is divided into

two time periods: peak-hours (6PM-0AM) and off-peak hours (0AM-6PM). In the used data sets, a maximum of 60,000 concurrent users on a single channel occured during peak hours.
In other research studies, P2P networks were also simulated to analyze the network behaviour [14; 2; 12]. However, the results from those simulated networks are not always reliable as they are not able to represent all the interferences which occur in the real internet [13].
In the Zattoo measurement study, sharing ratio, overlay size, channel switching delay as well as percentage of packet retransmissions were measured. Sub-stream and peer synchrony as well as the efficiency of error correction were measured on client-side [3]. Client-side measurements give mainly information about the quality of service experience by the users rather than network scalability. For measurement on client side, in the Zattoo Network study, a specialized Zattoo client was used to determine the internal states of other active Zattoo Peers. This leads to more realistic results. In study [14], NAT traversal issues were analyzed more deeply.

3.1 Channel switching delay

The channel switching delay is defined as the required time between the when user selects a channel and when he can actually watch the streamed video. This time span contains the network joining delay and the media prebuffer duration.
The join delay mainly depends on the used P2P network, sharing ratio and number of peers in the network and was measured by evaluating the feedback logs submitted by the users. However, the feedback logs are not fully representative as they are only submitted when certain errors occur during playback. Hence, the feedback logs are also compared to the session logs.
The length of the join delay also depends on whether live streaming is performed during or outside the peak hours. According to the measurements done in [3], during peak hours, the median join delay increases by 22%. The results also reveal that 80% of the users experience a join delay up to 8 seconds and 50% of the users have join delay with less than 2 seconds.
The prebuffer duration depends on the prevailing network conditions and how the live streaming is encoded. To decrease the prebuffer duration while assuring flawless playback, elaborate heuristics for bitrate adaptation are required. However, in [3] no special measurements regarding prebuffer delay were performed.
Usually, the channel switching delay has an overall length of about 10 to 15 seconds. In extreme cases, it can take up to 2 minutes before the user is able to watch the desired channel [14].

3.2 Sharing ratio and overlay size

The session log protocols contain timestamps when a peer joined and left the P2P delivery network. With those join/leave timestamps, the number of concurrent users on a given channel at a certain point in time can be determined. The session database also contains information about the upload and download bandwidth of the peers. Based on this information and by assuming that the amount of uploaded and downloaded bytes is equally distributed throughout the entire session duration, the sharing ratio can be calculated by dividing amount of uploaded bytes by the amount of downloaded bytes.
The average sharing ratio in the Zattoo network is between 0.22 and 0.34 which translates to a peer upload bandwidth from 100 kbps to 175 kbps. In Spain, the sharing ratio is lower on average because the upload bandwidth per peer is lower on average than in Germany and Switzerland.
The sharing ratio, upload capacity and session length are important parameters as they

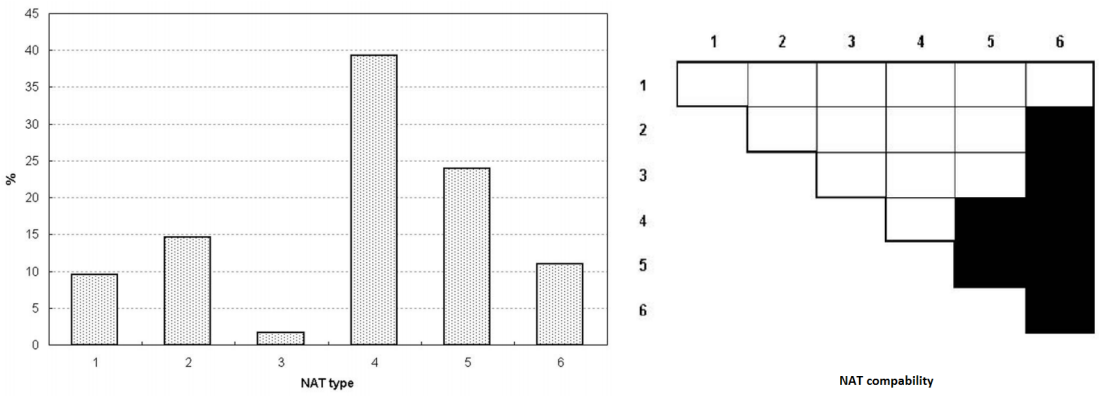

Figure 2: This figure shows the distribution and the compability of the NAT types of the peers. Most peers have a NAT type of 4 or 5. Some peers have a NAT type of 1, 2, or 6. There is a negligible amount of peers having a NAT type of 3. The compability of these types to eachother is shown on the right side. *Source: [14]*

determine the scalability of a P2P network [14]. According to [14], only 18% of the peers are able to redistribute more than the full stream. The sharing ratio k also determines the maximum number of total peers n in the network [14]:

- If $k > 1$, n does not converge and scales exponentially.
- If the sharing ratio $k = 1.0$, the n scales linearly.
- If the sharing ratio k of the peers is less than 1.0, n converges to $\frac{C}{1-k}$, where C is the number of peers which can directly contact the source peer (i.e. the encoding server). Thus, the system does not scale and the P2P network will be dependent on super node peers. This leads to a decreased network scalability in user size and an increased centralization as the super nodes are mostly controlled and provided by the P2P provider.

Besides sharing ratio, there are several other factors which determine the network scalability such as operational cost, the peer division multiplexing and network topology. Another important factor is the NAT compability of a peer. This issue will be discussed in the following section.

3.3 NAT traversal

Most networks use some type of NAT (Network Address Translation), which complicates peer-to-peer connections. NAT modifies IP address information in IP packet headers while traversing a traffic routing device. The NAT traversal problem occurs when two peers behind distinct NATs try to communicate. One way to solve this problem is to use port forwarding, another way is to use various NAT traversal techniques [14].
The connection between two peers can be done with either TCP or UDP. TCP was originally designed for transmissions which do not tolerate any data loss and not for real-time media delivery [7]. Hence, sent packets are always re-sent increasing the delay time and thus, a real-time playback cannot be guaranteed. However, TCP is less affected by NAT traversal issues.
Mostly, live streaming is performed with UDP because a real-time delivery of the media can be assured and lost packets do not always need to be retransmitted. In this term paper, only the peer communication over UDP will be considered. This makes UDP more

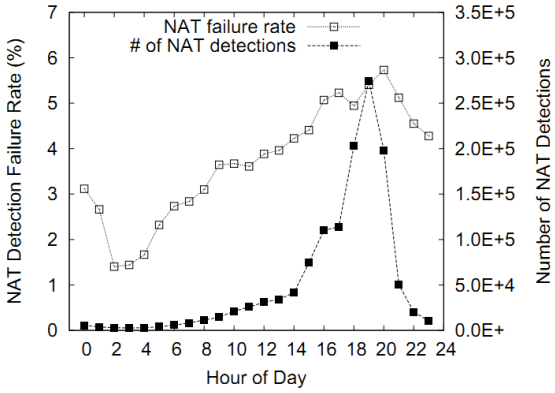

Figure 3: This figure shows the amount and failure rate of NAT detections. *Source: [3]*

suitable for video streaming. However, the main problem of UDP is the lack of availability because of NAT traversal issues.
To allow communication between NATed peers, their NAT configuration needs to be identified. Zattoo uses a UDP-based standard STUN-protocol[1] to perform the NAT detection. The STUN-protocol communicates with external STUN-servers to discover the presence and the type of the NAT gateway a peer utilizes . The Zattoo P2PTV system supports 6 different NAT configurations assigned with a number from 1 (the least restrictive type) to 6 (the most restrictive type): open host (1), full cone (2), IP-restricted (3), port-restricted (4), symmetric (5), and UDP-disabled (6) [14]. The NAT compability of peers with different NAT types is shown in figure 2. NAT incompabilities cause communication blocks between two peers.
The session database stores information about public and private IP adress as well as NAT type and configuration for each session. This information is used to analyze the NAT traversal behaviour of the peers. If the public/private IP adress pair leads to different NAT configurations, a NAT detection failure occured. By comparing the public/private IP adress pair with the associated NAT configurations, the failure rate of NAT detection can be determined.

In figure 3 the daily trend of NAT detection failure rate derived from the results of NAT detection is plotted. The NAT detection failure rate at hour x is the number of failed NAT detection results divided by the total number of NAT detections occurring on that hour. The total number of NAT detections also indicates how busy Zattoo system was at a certain point in time. The NAT detection failure rate is around 1.5% at 2-3AM in off-peak hours. During the busiest times at 8PM during the peak hours, the NAT type of about 6% of clients is not determinable which means that they do not contribute any bandwidth to the P2P network.
In [14] the effects NAT compability were examined in a simulated P2P network. An open-host and a NATed network were simulated. NAT incompabilities decrease the overall bandwidth in the network to roughly 15%. Different NAT types also have a different sharing performance. The symmetric NAT (NAT type 3) has the lowest sharing ratio and is however the most occuring NAT type in the Zattoo Network, as it can be seen in figure 2. Therefore, symmetric NATs can significantly affect the overall network sharing performance [3].
To examine how different NAT types affect the sharing ratio, in study [3] an experiment was performed with four Zattoo clients with four different and most occuring NAT types.

[1]STUN = Session Traversal Utilities for NAT. For a description of the functionality of the STUN-protocol, the reader is referred to [8].

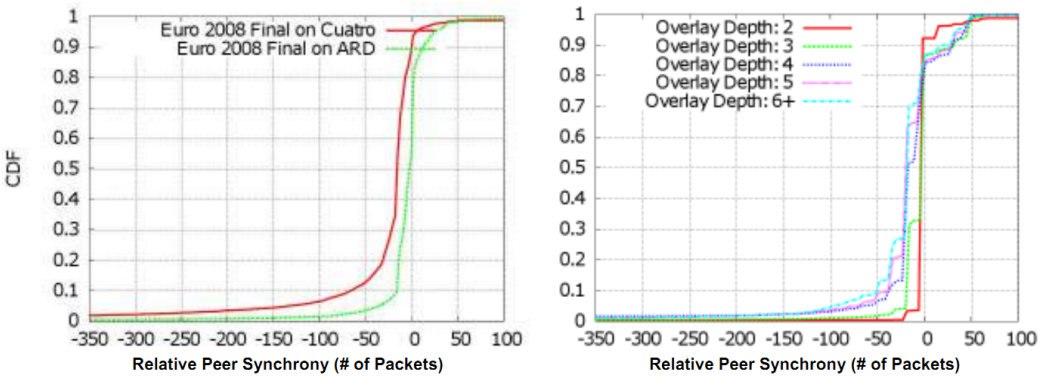

Figure 4: The CDFs of relative peer synchrony on different channels (left) and compared to the depth of the overlay network (right). Most of the peers have a delay of about 60 packets which is around one second in a 500 kbps stream. In the right diagram, it can be seen that the peer synchrony is decreasing with increasing overlay size. *Source: [3]*

The clients ran concurrently about 30 minutes and for better reliability, the tests were repeated three times. Afterwards, for each client the upload capacity and the average upload bandwidth utilization were measured. The symmetric NAT only utilized 20% of the upload bandwidth while the full-cone NAT used nearly 100%.

3.4 Sub-stream and peer synchrony

The approach for constructing the PDM in Zattoo favors sub-streams that are received synchronously. Out-of-sync received sub-streams result in a pixelated screen on the player. The sub-stream synchrony was measured by the difference between requested and received sub-streams. It can determined by analyzing the SEARCH replies from existing peers out of the session log files. A SEARCH reply contains the minimum and maximum packet sequence numbers of received sub-streams and information about the status and correctness of those sub-stream packets. In the measurements of the Zattoo network, about 99% (ARD) and 96% (Cuatro) have 3 or less bad sub-streams. This means that the number of bad sub-streams is not high enough to cause any quality issues.

The sub-stream synchrony provides information about the video quality whereas the peer synchrony affects the time shift of the viewing point between different peers.
The synchrony decreases with network scale and distance of a peer between encoding servers. Hence, within small P2P networks, most of the peers watch video relatively synchronously in time whereas in large networks the time shift can differ significantly between the peers. The viewing point of a peer is defined as the median of latest packet number of the sub-streams. As a reference point, a peer very close the encoding server is selected. Afterwards, its viewing point is compared to all other peers.
The peer synchrony was measured in packets. A peer synchrony of -60 packets means that the viewing point is delayed by 60 packets which is roughly one second for a 500 kbps stream. Peers directly connected to an encoding server often have a positive peer synchrony. Figure 4 shows the results in form of CDFs obtained from the measurements done in [3].
If peers have a larger distance to the encoding server, a left shift of the CDFs is expected. The median delay at depth 6 does not by more than a half second compared to the median delay at depth 2. Figure 4 also shows that the number of peers lagging 0.5 seconds behind the reference viewing increases with growing overlay depth. Even for a high overlay depth, the delay caused by additional peer hops is low enough to achieve real-time conditions.

4 Discussion and conclusion

This term paper described the functionality of a P2P live streaming system and presented the obtained results of the studies [3; 14]. The studies analyzing large amounts of usage data collected in form of session and user feedback log files during one of the largest viewing event in Europe. It was shown that the Zattoo network can scale to a large number of users although the user peers upload bandwidth is often limited.
However, the performed measurements showed that the functionality of the Zattoo network can also be improved in several directions. The main problems are limited upload bandwidth and NAT traversal of user peers, channel switching delay and synchrony of peers.
Due to the limited upload bandwidth of user peers which is often less than the compressed playback rate of the video, super-nodes and repeater nodes are still required for streaming of high definition video. Those peers with a high uploading capacity upload much more than what they download to support the continous playback at all peers. This leads to an unequal sharing ratio. Possible solutions to be considered are increasing number of upload/download symmetric broadband connections (for example VDSL with fiber optic cables) and to provide more super-nodes and repeater-nodes.
Another issue affecting the overall sharing ratio of the peers is peer NAT configuration. The Zattoo network performs a NAT configuration detection for each peer. However and most notably during the peak hours, this NAT detection fails at up to 6% of the peers leading them to not contribute any bandwidth to the network. Thus, better NAT detection and traversal schemes need to be explored. One solution is the ICE [2] procedure. It allows for establishing connectivity even in the presence of NATs by using STUN/TURN servers. ICE provides a framework for dealing with NAT traversal and detection problems [11]. Currently, it is still in its design phase by the IETF. Fortunately, with the upcoming of the new IPv6, the presence of NATs will become a minor problem [5].
The Zattoo system and other established P2P streaming networks use variable-bit rate streams to achieve a better quality-to-bandwidth ratio compared to constant-bit rate encoding. The presence of variable-bit rate streams requires the peers for to perform a kind of rate adaptation which considers the prevailing network condition and the available up- and download bandwidth of the peer. Furthermore, layered video with SVC may lead to better bandwidth utilization [12].
In paper [1], it was measured that some peers watch frames in a channel minutes behind other peers. In contrast, study [3] revealed that the influece of number of peer hops can be neglected and the current PDM approach of [3] is already able to run a hybrid P2P/CDN delivery network with repeater nodes. Furthermore, better strategies for building the overlay network need to be explored to minimize the number of hops between the peers and the number of required repeater nodes.
Many users of ordinary television are used to switch rapidly between different channels. Hence, the channel switching delay should be as low as possible. Thus, if P2P live streaming systems establish to be a replacement for ordinary television, the channel switching delay needs to be reduced to from tens of seconds to a few seconds or even fractions of seconds. A possible solution would be to stream more than one channel at a time. However, this also means that the download traffic is increased and a higher upload bandwidth is required. Additionally, a better coding of the video segments may shorten the delay times. Other solutions have to consider shorter buffering times with better rate adaptation approachs and decreasing of the time required for P2P network joining phase.
P2P live streaming is still a marginal phenomenon compared to digital satellite TV broadcasting solutions due to quality of service and network scalability issues in terms of channel switch time and limited peer upload bandwidth. The obtained results can be useful for architecting and improving the performance of a large scale P2P live streaming system. With increasing peer upload bandwidths and the emerge of the IPv6, the future for P2P live streaming is promising.

[2] Interactive Connectivity Establishment; RFC 5245 [9]

References

[1] Shahzad Ali, Anket Mathur, and Hui Zhang, *Measurement of Commercial Peer-To-Peer Live Video Streaming*, Tech. report, School of Computer Science Carnegie Mellon University, 2006.

[2] Suman Banerjee, Christopher Kommareddy, Koushik Kar, Bobby Bhattacharjee, and Samir Khuller, *Construction of an Efficient Overlay Multicast Infrastructure for Real-time Applications*, Tech. report, Department of Computer Science, University of Maryland, 2003.

[3] Hyunseok Chang, Sugih Jamin, and Wenjie Wang, *Live Streaming Performance of the Zattoo Network*, Tech. report, Alcatel-Lucent Bell Labs, 2008.

[4] Cisco White Paper, *Cisco Visual Networking Index: Forecast and Methodology, 2009–2014*, June 2010, `http://www.cisco.com/en/US/solutions/collateral/ns341/ns525/ns537/ns705/ns827/ white_paper_c11-481360.pdf`.

[5] Kaushik Das, *Network Address Translation (NAT): Pros and Cons*, IPv6.com (2008), `http://www.ipv6.com/articles/nat/NAT-Pros-and-Cons.htm`.

[6] Xiaojun Hei, Yong Liuz, and Keith W. Rossy, *Inferring Network-Wide Quality in P2P Live Streaming Systems*, Tech. report, Polytechnic University, Brooklyn, NY, USA, 2007.

[7] The Internet Society, *Transmission Control Protocol*, September 1981, `http://tools.ietf.org/html/rfc793`.

[8] The Internet Society, *Session Traversal Utilities for NAT (STUN)*, October 2008, `http://tools.ietf.org/html/rfc5389`.

[9] The Internet Society, *Interactive Connectivity Establishment (ICE): A Protocol for Network Address Translator (NAT) Traversal for Offer/Answer Protocols*, April 2010, `http://tools.ietf.org/html/rfc5245`.

[10] Gunnar Kreitz and Fredrik Niemelä, *Large Scale, Low Latency, P2P Music-on-Demand Streaming*, Tech. report, KTH – Royal Institute of Technology, and Spotify Stockholm, Sweden, 2010.

[11] Patrik Persson, Xing Fan, Yuan Song, and Stefan Håkansson, *Beyond HTML5: P2P Conversional Video*, Tech. report, Ericsson Labs, 2011, `https://labs.ericsson.com/developer-community/blog/beyond-html5-peer-peer-conversational-video`.

[12] Thomas Schierl, Yago Sanchez, Cornelius Hellge, and Thomas Wiegand, *Improving P2P Live-Content Delivery using SVC*, Tech. report, Fraunhofer HHI, Image Processing Department, Germany, 2010.

[13] Mario T. Schlosser, Tyson E. Condie, and Sepandar D. Kamvar, *Simulating a File-Sharing P2P Network*, Tech. report, Department of Computer Science Stanford University, Stanford, USA, 2004.

[14] Khaldoon Shami, Damien Magoni, Hyunseok Chang, Wenjie Wang, and Sugih Jamin, *Impacts of Peer Characteristics on P2PTV Networks Scalabilty*, Tech. report, University of Haute Alsace, Colmar, France, 2009.